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Potato Leaf Disease Classification Using MobileNetV3Small Transfer Learning with Dataset Split Ratio Comparison

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Abstract

Early blight and late blight substantially reduce potato yield when diagnosis is delayed, while manual identification remains slow, expertise-dependent, and subjective. This study developed a MobileNetV3Small transfer-learning classifier and examined how three stratified training–validation–test partitions—60:20:20, 70:15:15, and 80:10:10—affected predictive performance. A dataset of 4,988 images from Early Blight, Healthy, and Late Blight classes was resized to 224×224 pixels, normalized, and augmented only within the training subset. The model was trained through feature extraction and fine-tuning, with each partition repeated three times. These repetitions simultaneously varied dataset allocation, parameter initialization, mini-batch construction, and stochastic augmentation; therefore, the reported variability reflects combined run-to-run stochastic effects rather than split-ratio stability alone. Across repetitions, the 70:15:15 partition achieved mean±SD accuracy, weighted precision, weighted recall, and weighted F1-score of 96.53±0.35%, 96.59±0.34%, 96.54±0.31%, and 96.50±0.36%, respectively. Corresponding values were 96.09±0.46%, 96.20±0.41%, 96.12±0.49%, and 96.09±0.47% for 60:20:20, and 95.79±0.80%, 95.97±0.70%, 95.79±0.79%, and 95.77±0.80% for 80:10:10. The highest single-run accuracy was 96.80% under 70:15:15, with weighted precision, recall, and F1-score of 96.85%, 96.78%, and 96.78%. Class-specific evaluation nevertheless showed lower Late Blight recall of approximately 93–94%. Under these specific experimental conditions, 70:15:15 provided the most favourable balance of performance and consistency, without implying general optimality across datasets or architectures. External datasets, controlled repeated experiments, and independent external testing are required before broader recommendations can be made.

I. INTRODUCTION

Potato (Solanum tuberosum L.) is one of the most strategically important horticultural commodities in Indonesia, where domestic consumption and demand from the food-processing industry have driven a steady rise in national production over recent years [1]. This economic value, however, is persistently threatened by foliar diseases, most notably early blight and late blight, which spread rapidly across a plantation and can cause severe yield losses once an outbreak is left unmanaged [2], [3]. In most smallholder settings the diseases are still identified visually by farmers or field officers, a practice that is inherently slow, requires specialised agronomic knowledge, and varies from one observer to another. Under the humid highland climate typical of Indonesian potato cultivation, even a short delay in diagnosis can turn a manageable infection into a destroyed harvest, which makes timely and objective detection a matter of real economic consequence.

Advances in artificial intelligence, and convolutional neural networks (CNN) in particular, have been widely adopted to overcome the limitations of manual inspection because these networks learn discriminative visual patterns directly from leaf images in an automatic and reproducible manner [4], [5]. Systematic reviews of the

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field confirm that CNN-based pipelines now dominate plant-disease recognition and that the research frontier has shifted from heavy, high-parameter backbones toward lightweight architectures suited to on-device deployment [6]. The availability of open image repositories such as PlantVillage has further accelerated this progress by providing tens of thousands of freely accessible healthy and diseased leaf images for experimentation [7]. Training a CNN from scratch, nevertheless, generally demands a large annotated corpus and a considerable computational budget, resources that are rarely available in typical agricultural research settings.

Transfer learning offers a pragmatic remedy to this constraint, since it allows a model to reuse representations learned previously from large-scale datasets such as ImageNet and then adapt them to a new, smaller classification problem [8], [9]. Among the many pre-trained backbones, MobileNetV3Small is especially attractive for agricultural applications because it was explicitly designed—through neural architecture search combined with inverted-residual bottlenecks and squeeze-and-excitation attention—to remain lightweight and computationally efficient without a substantial accuracy penalty [10]. This efficiency makes it a realistic candidate for the resource-constrained smartphones that farmers already carry into the field, where a compact model can support early, on-site screening of suspected infections without dependence on cloud connectivity.

Several studies have already reported encouraging results for transfer-learning-based potato leaf disease classification. A stacking-ensemble approach that combines multiple transfer-learning backbones improved classification accuracy relative to any single model, albeit at a markedly higher computational cost [11]. A recent enhancement of MobileNetV3Small that fuses additional feature-extraction modules likewise raised accuracy on potato lesions, confirming that the base architecture is a competitive starting point that still leaves room for refinement [12]. In parallel, image augmentation techniques such as rotation, flipping, contrast variation, and noise injection have been shown to strengthen a model's robustness to field-condition variability while reducing the risk of overfitting on limited training data [13].

Despite this progress, most existing work concentrates on designing or modifying the model architecture, whereas the way the dataset is partitioned into training, validation, and test subsets is seldom examined as a variable that materially affects performance. Evidence from adjacent domains is instructive: experiments on custom image datasets have demonstrated that changing the train/test proportion can shift the accuracy of pre-trained models by a non-trivial margin [14], and studies in medical image processing have reported a clear trade-off between the training and testing ratio that influences the stability of a model's generalisation [15]. These observations imply that the split proportion should not be chosen arbitrarily but must be validated empirically for each classification task, including potato leaf disease classification, which has rarely been analysed from this perspective.

Building on this gap, the present study develops an augmented MobileNetV3Small classifier for potato leaf disease and, as its central contribution, quantifies the effect of three dataset split scenarios—60:20:20, 70:15:15, and 80:10:10—on the resulting performance. Each scenario is trained across three repetitions so that the reported outcomes reflect not only peak accuracy but also consistency between runs, a dimension frequently overlooked in single-run studies. The study therefore aims to recommend the split proportion that is simultaneously the most accurate and the most stable for comparable lightweight plant-disease image-classification research.

II. Literature Review

Research on potato leaf disease diagnosis has progressed from application-oriented systems to increasingly sophisticated classification pipelines. A lightweight disease-detection application showed that machine-learning support can be implemented in a practical interface [1]. Subsequent work extended the task to disease identification and severity assessment [2]. Feature-selection methods combined with optimized classifiers have also achieved competitive results while reducing representation dimensionality [3]. However, these studies primarily represent model-development research: they differ in dataset provenance, class composition, sample size, feature representation, and evaluation protocol, while emphasizing application functionality, severity estimation, or feature optimization. Their results are therefore not directly comparable, and they do not establish how alternative allocations of the same dataset affect performance estimates.

Transfer learning provides a practical basis for image classification when domain-specific labelled data are limited because it adapts representations learned from a large source dataset to a smaller target task [8]. Convolutional neural networks progressively encode edges, textures, shapes, colour changes, and lesion patterns [9]. MobileNetV3 was designed to balance predictive capacity and computational efficiency through neural architecture search and mobile-oriented principles [10]. Nevertheless, architectural efficiency alone does not demonstrate diagnostic robustness. Performance also depends on image provenance, background conditions, the number of independent plants represented, and whether evaluation images are genuinely distinct from training images. Lightweight deployment potential must therefore be distinguished from evidence of real-field generalization.

Potato-leaf studies have increasingly pursued architectural complexity. A stacking ensemble combining several pretrained backbones can exploit complementary representations and achieve strong performance [11]. However, it increases memory use, inference time, and implementation complexity. An enhanced

MobileNetV3Small subsequently introduced additional feature-extraction modules to improve disease recognition [12]. Although such modifications may raise accuracy, they make it difficult to separate gains caused by architecture from those caused by data preparation or evaluation design. Unlike these model-development studies, the present research keeps the MobileNetV3Small architecture and training procedure constant while examining data allocation as the main experimental factor.

Image augmentation increases training diversity through rotation, flipping, translation, zooming, brightness and contrast adjustment, and noise injection [13]. Yet augmentation cannot correct weaknesses in partition design. Random image-level splitting may preserve class proportions while allowing duplicate images, repeated views of the same leaf, or source-specific background artefacts to cross training, validation, and test subsets. High accuracy may then partly reflect visual similarity or background leakage rather than robust disease recognition. Stratification addresses class balance but not sample independence; when provenance metadata are available, group-aware splitting by leaf, plant, acquisition session, field, or source is preferable. Experiments on custom image datasets show that changing the train-test proportion can materially alter pretrained-model performance [14]. Medical-image research similarly demonstrates a trade-off between allocating enough observations for learning and retaining a sufficiently large test set for reliable evaluation [15]. These findings support explicit partition analysis but also highlight the importance of repeated holdout validation, learning curves, confidence intervals, and protection against model-selection bias.

MobileNetV3Small builds on inverted residuals and linear bottlenecks introduced in MobileNetV2 [16] and squeeze-and-excitation mechanisms that recalibrate informative channels [17]. An optimized MobileNetV2 model has shown that compact transfer-learning architectures can achieve high accuracy on potato-leaf images [18]. Related studies have evaluated MobileNetV3-Small for rice diseases [19], combined EfficientNetV2 with hyperparameter tuning [20], and developed extremely lightweight MobileNetV2 variants [21]. Other approaches include Twin-CNN feature fusion [22], privacy-preserving detection [23], and CNN classifiers for paddy diseases [24], [25]. Collectively, these studies demonstrate architectural diversity but seldom report dataset provenance, duplicate control, group-aware partitioning, repeated validation, external testing, or test-set uncertainty consistently enough for fair comparison.

The research gap is therefore not the general discovery that split proportions influence performance, but the limited controlled evidence on split-ratio sensitivity for a fixed lightweight architecture on one potato-leaf dataset. This study compares stratified 60:20:20, 70:15:15, and 80:10:10 allocations under an identical MobileNetV3Small configuration and repeats each scenario three times. It evaluates mean performance and run-to-run dispersion while recognizing that data allocation, parameter initialization, mini-batch construction, and stochastic augmentation vary jointly. The contribution is consequently a dataset- and protocol-specific empirical comparison rather than a universal partitioning rule, and its conclusions remain bounded by random image-level splitting and the absence of independent external validation.

III. METHODS

A. Research Design and Workflow

This study employed an experimental image-classification design to investigate the effect of dataset split ratios on the performance of a MobileNetV3Small-based potato leaf disease classifier. The research procedure was organised as a sequential pipeline consisting of eight main stages, namely dataset loading, image preprocessing, dataset splitting, image augmentation, model building, model training, model evaluation, and result comparison. The complete sequence of these stages is presented in Figure 1.

The dataset split ratio was treated as the principal experimental factor in this study. Three training, validation, and testing configurations were examined, consisting of 60:20:20, 70:15:15, and 80:10:10. To ensure a controlled comparison, the preprocessing procedure, augmentation strategy, model architecture, transfer-learning mechanism, and evaluation metrics were maintained consistently across all experimental scenarios.

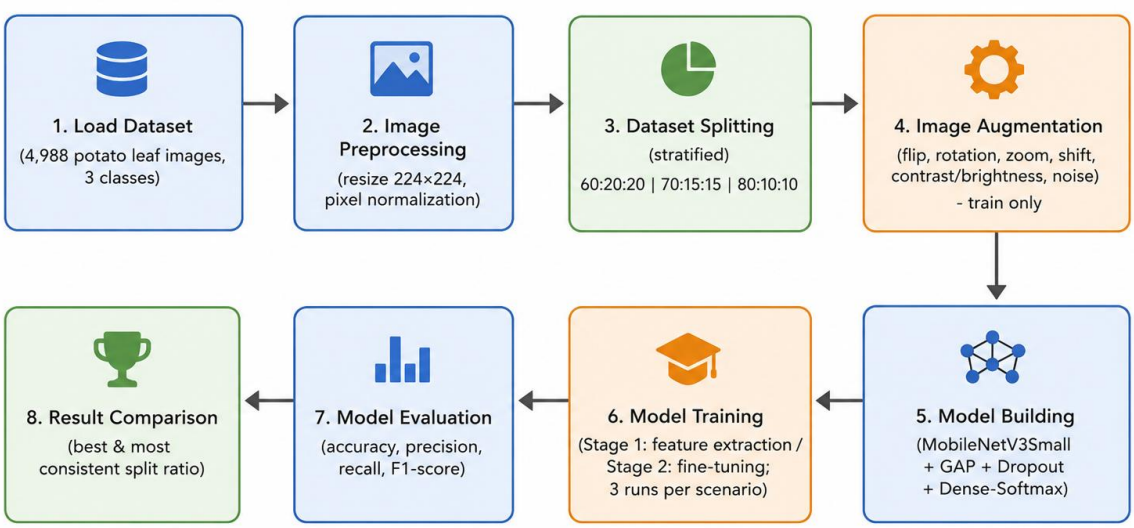


Fig. 1 Flowchart of the Research Method

B. Dataset Loading

The first stage involved loading all potato leaf images and their corresponding labels from directories organised according to the three predefined categories. The dataset comprised 4,988 images, including 1,828 Early Blight images, 1,624 Late Blight images, and 1,536 Healthy images, representing approximately 36.65%, 32.56%, and 30.79% of the dataset, respectively. Each image was assigned a target label based on the name of its parent directory, thereby establishing a consistent mapping between the image file and its corresponding disease category.

Although the number of images was not identical across the three classes, the class distribution was sufficiently comparable, and no single class overwhelmingly dominated the dataset. Examining class distribution before model training is important because substantial class imbalance may bias a classifier toward the majority class and produce performance estimates that do not adequately represent minority-class recognition [26]. Therefore, stratified partitioning was subsequently employed to preserve approximately the same class proportions within the training, validation, and testing subsets, thereby reducing distributional differences among the experimental partitions and supporting a more reliable comparison of the dataset split scenarios.

C. Image Preprocessing

The second stage standardised the input images according to the input specifications of the MobileNetV3Small architecture. Each image was resized to 224×224 pixels to ensure a uniform spatial dimension and consistent processing by the pre-trained network. This standardisation is essential in convolutional neural network pipelines because a fixed input shape is required for reliable feature extraction and classification [9], [10].

After resizing, the pixel values were normalised using the preprocessing procedure recommended for MobileNetV3Small so that they matched the numerical range expected by the pre-trained ImageNet weights. This process reduced scale differences, supported more stable gradient-based optimisation, and maintained consistency between the target images and the data representation used during pre-training [8], [10]. The same resizing and normalisation procedures were applied to the training, validation, and testing subsets to ensure that performance differences could be attributed primarily to the dataset split ratio rather than to inconsistencies in data preparation.

D. Dataset Splitting

The third stage divided the dataset into training, validation, and testing subsets using random stratified partitioning. Stratification was applied to preserve approximately the same proportions of Early Blight, Late Blight, and Healthy images in each subset, thereby reducing the risk that differences in class distribution would bias the experimental results. The training subset was used to estimate and update the model parameters, the validation subset was used to monitor generalisation performance and support model selection during training, and the testing subset was kept separate from the learning process for final performance evaluation.

Three training-validation-testing configurations were examined, namely 60:20:20, 70:15:15, and 80:10:10. Comparing these scenarios enabled the study to investigate the trade-off between allocating more samples for model learning and retaining sufficiently large validation and testing subsets for reliable performance estimation. Previous studies have shown that variations in train-test proportions can affect both the predictive performance and stability of pre-trained image-classification models, particularly when smaller evaluation subsets become more sensitive to the specific samples selected [14], [15].

E. Image Augmentation

Image augmentation was applied exclusively to the training subset to increase the visual diversity of the available samples while preserving their original class labels. This strategy synthetically generated plausible image variations, thereby reducing the model's dependence on limited visual patterns and lowering the risk of overfitting during training. Data augmentation is widely used in image classification because it can improve model robustness and generalisation without requiring additional manually labelled images [13].

The augmentation operations included horizontal and vertical flipping, random rotation, zooming, translation, contrast adjustment, brightness modification, and mild noise injection [13]. These transformations were randomly applied during training to simulate variations in leaf orientation, camera distance, illumination, positioning, and image quality that may occur during image acquisition. No augmentation was performed on the validation and testing subsets, which were retained in their original form to ensure that model selection and final evaluation were conducted using previously unseen and unmodified images, while also preventing augmented derivatives of training samples from introducing data leakage into the evaluation process.

F. MobileNetV3Small Model Architecture

The fifth stage constructed the classification model using MobileNetV3Small pre-trained on the ImageNet dataset as the convolutional feature-extraction backbone [10]. The architecture was selected because it was designed to provide a balance between predictive performance and computational efficiency. Its lightweight characteristics also make it relevant for future implementation on mobile or resource-constrained devices.

The original classification layer of MobileNetV3Small was replaced with a task-specific classification head. This classification head consisted of a global average pooling layer, a dropout layer, and a fully connected dense layer. The global average pooling layer transformed the convolutional feature maps into a compact feature vector, while the dropout layer was used to reduce the risk of overfitting.

The final dense layer contained three output units corresponding to the Early Blight, Late Blight, and Healthy classes. A softmax activation function was applied to convert the model outputs into probability scores for the three classes. The class with the highest probability value was selected as the final prediction.

G. Model Training

The model was trained using a two-phase transfer-learning strategy consisting of feature extraction and fine-tuning. During the feature-extraction phase, all layers of the MobileNetV3Small backbone were frozen, while only the newly added classification head was trained using the potato leaf images. This approach allowed the classifier to reuse the general visual representations learned from the large-scale ImageNet dataset, thereby reducing the computational requirements and data demands associated with training a convolutional neural network from scratch [8], [10].

In the second phase, selected upper layers of the MobileNetV3Small backbone were unfrozen and trained jointly with the classification head using a lower learning rate. Fine-tuning the higher-level layers enabled the pre-trained representations to adapt gradually to domain-specific characteristics of potato leaf images, including lesion colour, texture, shape, and spatial pattern, while minimising the risk of substantially altering useful low-level features such as edges and contours [8], [9]. The same feature-extraction and fine-tuning procedures were consistently applied to all dataset split scenarios so that differences in performance could be attributed primarily to the dataset partition rather than to variations in the training configuration.

Each dataset split scenario was trained three times using different random seeds to account for stochastic variation arising from data partitioning, parameter initialisation, mini-batch construction, and augmentation operations. Repeated training runs provided a more reliable assessment of model performance than a single experiment because they enabled the consistency of the resulting accuracy, precision, recall, and F1-score values to be examined across different training conditions. The mean performance and standard deviation across the three repetitions were subsequently considered when comparing the split scenarios, thereby reducing dependence on an unusually favourable or unfavourable individual run.

H. Model Evaluation

The seventh stage evaluated each trained model using the previously unseen testing subset. Performance was measured using accuracy, precision, recall, and F1-score. These metrics were selected because accuracy alone may not provide a complete explanation of model behaviour across multiple classes.

Accuracy represents the proportion of correctly classified images among all test samples. Precision measures the proportion of correct positive predictions among all samples predicted as belonging to a particular class. Recall measures the proportion of actual samples from a class that are successfully identified by the model.

Accuracy was calculated using the following equation 1:

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

Precision was calculated using the following equation 2:

$$\text{Precision} = \frac{TP}{TP+FP} \quad (2)$$

Recall was calculated using the following equation 3:

$$\text{Recall} = \frac{TP}{TP+FN} \quad (3)$$

The F1-score was calculated as the harmonic mean of precision and recall. This metric provides a balanced evaluation when both false-positive and false-negative errors need to be considered. The F1-score was calculated using the following equation 4:

$$\text{F1-score} = 2\left(\frac{\text{Precision}\times\text{Recall}}{\text{Precision}+\text{Recall}}\right) \quad (4)$$

In these equations, TPdenotes true positives, TNdenotes true negatives, FPdenotes false positives, and FNdenotes false negatives. For the multiclass classification task, these values were determined for each disease class based on the predicted and actual labels. The aggregate precision, recall, and F1-score values were then used to compare the overall performance of the experimental scenarios.

A confusion matrix and classification report were also examined for the best-performing run of each dataset split scenario. The confusion matrix was used to identify the distribution of correct and incorrect predictions among the Early Blight, Late Blight, and Healthy classes. The classification report provided class-specific precision, recall, and F1-score values to reveal performance patterns that could not be identified through overall accuracy alone.

Validation accuracy was recorded as supplementary information during the model-training process. This metric was used to observe the model's generalisation behaviour on data that were not used to update the model parameters. Nevertheless, the primary comparison between split scenarios was based on performance obtained from the independent testing subset.

I. Reult Comparison

The eighth and final stage compared the outcomes obtained from the three dataset split scenarios. For each scenario, the results of the three training repetitions were summarised using the best accuracy, mean accuracy, standard deviation of accuracy, validation accuracy from the best-performing run, and best F1-score. These statistics enabled the scenarios to be assessed not only in terms of their highest predictive performance but also according to their overall consistency across repeated experiments.

Mean accuracy was used to represent the average performance achieved across the three repetitions, whereas the standard deviation quantified the degree of variation between runs. A lower standard deviation indicated that the model produced more stable results under different random seeds, data partitions, and stochastic training conditions. The best accuracy and F1-score were also reported to indicate the maximum classification performance achieved by each split configuration.

The preferred dataset split scenario was selected by jointly considering predictive performance and experimental stability rather than relying solely on the result of a single run. A scenario was regarded as more favourable when it achieved comparatively high mean and best test performance while maintaining a low standard deviation across repetitions. Accordingly, the final comparison identified the dataset partition that provided the most accurate and consistent performance under the experimental conditions of this study.

IV. Results

A. Performance Across Dataset Split Scenarios

The dataset comprised 4,988 potato leaf images distributed across the Early Blight, Late Blight, and Healthy classes. Because the class distribution was relatively balanced, stratified partitioning was employed to maintain approximately equivalent class proportions within the training, validation, and test subsets. MobileNetV3Small was trained using the same two-stage procedure—initial feature extraction followed by fine-tuning—under the 60:20:20, 70:15:15, and 80:10:10 split scenarios. Each scenario was independently repeated three times using different random seeds to evaluate both predictive performance and run-to-run stability.

The complete test-set results for all nine experimental runs are presented in Table 1. Across the three split scenarios, test accuracy ranged from 94.99% to 96.80%, indicating that MobileNetV3Small maintained a high level of classification performance under different data-allocation configurations. Nevertheless, the magnitude of variation across repetitions differed among the scenarios. The 70:15:15 scenario produced accuracy values

between 96.13% and 96.80%, whereas the 60:20:20 scenario ranged from 95.59% to 96.49%. The 80:10:10 scenario exhibited the widest range, from 94.99% to 96.59%.

TABLE 1
PER-RUN TEST RESULTS FOR EACH DATASET SPLIT SCENARIO

Scenario	Run	Accuracy	Precision	Recall	F1-Score	Val. Accuracy
60:20:20	1	0.9649	0.9662	0.9650	0.9650	0.9679
60:20:20	2	0.9559	0.9580	0.9557	0.9558	0.9770
60:20:20	3	0.9619	0.9618	0.9629	0.9619	0.9719
70:15:15	1	0.9666	0.9671	0.9664	0.9662	0.9853
70:15:15	2	0.9613	0.9621	0.9619	0.9610	0.9773
70:15:15	3	0.9680	0.9685	0.9678	0.9678	0.9786
80:10:10	1	0.9659	0.9668	0.9656	0.9657	0.9800
80:10:10	2	0.9499	0.9528	0.9499	0.9498	0.9719
80:10:10	3	0.9579	0.9596	0.9582	0.9576	0.9659

The 70:15:15 scenario produced the highest individual test performance in the third run, achieving an accuracy of 0.9680, precision of 0.9685, recall of 0.9678, and F1-score of 0.9678. The best result for the 80:10:10 scenario was obtained in the first run, with an accuracy of 0.9659 and an F1-score of 0.9657. Meanwhile, the best 60:20:20 configuration reached an accuracy of 0.9649 and an F1-score of 0.9650. Thus, the best run under the 70:15:15 split outperformed the strongest 80:10:10 and 60:20:20 runs by 0.21 and 0.31 percentage points in accuracy, respectively.

B. Aggregate Performance and Run-to-Run Stability

Because the differences between the best runs were relatively small, the scenarios were further evaluated using their mean performance and standard deviation across repetitions. Table 2 reports the mean and sample standard deviation for accuracy, precision, recall, and F1-score. This aggregated presentation reduces dependence on a single favourable run and provides a more informative assessment of experimental consistency.

TABLE 2
STATISTICAL SUMMARY OF TEST RESULTS PER SCENARIO

Scenario	Best Accuracy	Val. Acc. (best run)	Mean Accuracy	Std. Accuracy	Best F1-Score
70:15:15	0.9680	0.9786	0.9653	0.0035	0.9678
80:10:10	0.9659	0.9800	0.9579	0.0080	0.9657
60:20:20	0.9649	0.9679	0.9609	0.0046	0.9650

Values are reported as mean ± sample standard deviation across three independent runs.

As shown in Table 2, the 70:15:15 scenario achieved the highest mean values for all four test metrics. Its mean accuracy was 96.53%, compared with 96.09% for 60:20:20 and 95.79% for 80:10:10. The same ranking was observed for mean precision, recall, and F1-score, indicating that the advantage of the 70:15:15 scenario was not restricted to accuracy alone. Figure 2 presents the corresponding comparison of accuracy, precision, recall, and F1-score for the best-performing run under each scenario.

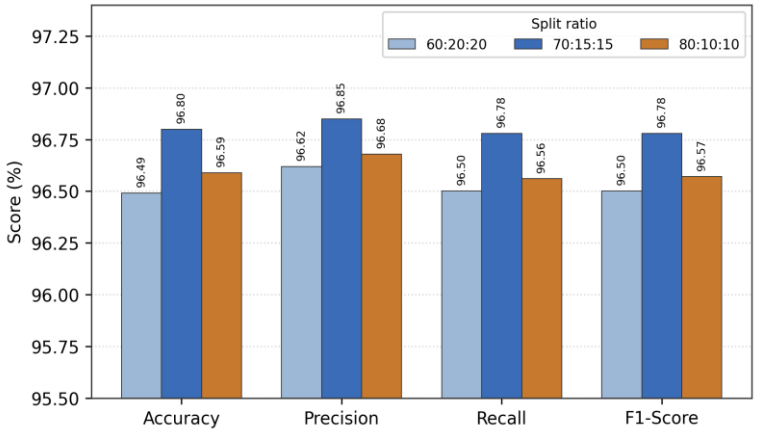


Fig. 2 Comparison of the best-run metrics for each dataset split scenario

The standard deviations provide additional evidence regarding the consistency of the three configurations. The 70:15:15 scenario recorded the lowest accuracy standard deviation at 0.35%, followed by 0.46% for 60:20:20 and 0.80% for 80:10:10. Its precision, recall, and F1-score also showed the lowest or nearly lowest variation across repetitions. In contrast, the 80:10:10 scenario demonstrated the greatest variability for all four test metrics,

including an accuracy range of 1.60 percentage points between its best and worst runs. The distribution of test accuracy across the three repetitions is illustrated in Figure 3.

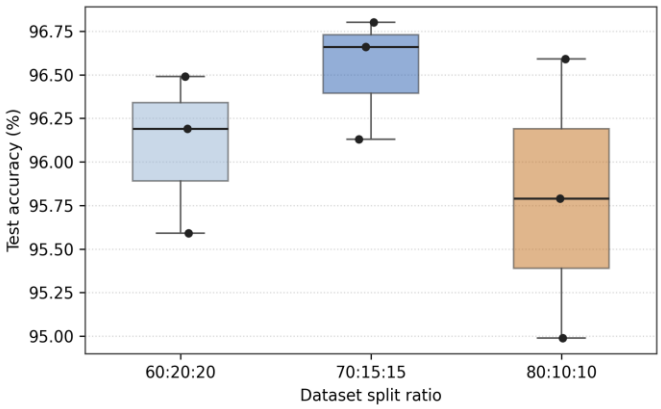


Fig. 3 Distribution of accuracy across three repetitions for each scenario

An additional distinction was observed between validation and test performance. Although the 80:10:10 scenario produced the highest individual validation accuracy of 98.00%, it obtained the lowest mean test accuracy and the largest test-accuracy standard deviation. Conversely, the 70:15:15 scenario did not produce the highest validation value but yielded the strongest mean test performance and the smallest run-to-run variation. These results demonstrate that the scenario ranking based on a single validation result was not identical to the ranking obtained from repeated test-set evaluation.

C. Class-Specific Classification Results

To examine whether the aggregate metrics concealed differences among disease categories, the classification reports and confusion matrices from the best-performing run of each scenario were analysed. Figure 4 shows that the Late Blight class consistently achieved the highest reported precision, reaching 1.00 in all three best-run models. At the reported level of precision, this result indicates that predictions assigned to the Late Blight category did not include false-positive classifications from the Early Blight or Healthy classes.

The recall for Late Blight was comparatively lower, ranging from approximately 0.93 to 0.94 across the three scenarios. Therefore, although the model produced highly precise Late Blight predictions, a small number of actual Late Blight samples remained assigned to another category. By comparison, the Early Blight and Healthy classes achieved recall values of approximately 0.98, indicating that most samples from these two classes were correctly identified. The remaining classification errors were therefore concentrated more strongly in false-negative Late Blight cases than in false-positive Late Blight predictions.

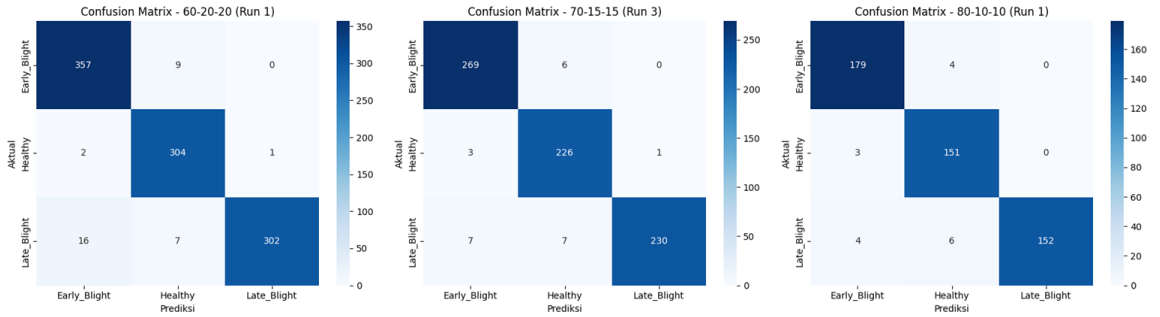


Fig. 4 Confusion Matrices of the Best Run In Each Scenario

Overall, the repeated experiments showed that all three split scenarios supported high classification performance, with the best-run accuracy exceeding 96% in every configuration. However, the 70:15:15 scenario provided the strongest combined result because it achieved the highest best-run accuracy, the highest mean accuracy, the highest mean precision, recall, and F1-score, and the lowest accuracy variation across repetitions. Under the experimental conditions examined, this configuration therefore provided the most favourable balance between predictive performance and empirical stability.

V. Discussion

The results indicate that MobileNetV3Small transfer learning can classify the three potato-leaf categories with high repository-level accuracy under all examined partitions, but this finding should be interpreted within the limits of the experimental setting. The model operated on relatively uniform images rather than independently

acquired field photographs; therefore, its performance reflects controlled image recognition rather than demonstrated agronomic diagnosis. The observed accuracy range is broadly consistent with potato-focused transfer-learning research: a stacking ensemble achieved strong predictive performance at the cost of greater architectural and computational complexity [11], whereas a modified MobileNetV3Small improved recognition through additional feature-extraction components [12]. The present model reaches a comparable accuracy band without ensembling or structural modification. Nevertheless, this study did not measure model size, parameter count, floating-point operations, training time, peak memory consumption, or inference latency on representative mobile or embedded hardware. MobileNetV3Small can therefore be described as architecturally lightweight, but an accuracy–efficiency trade-off and practical suitability for constrained devices have not yet been empirically demonstrated.

The split-ratio comparison provides descriptive evidence that 70:15:15 produced the most favourable observed balance between mean performance and run-to-run dispersion, but it does not establish substantive superiority. Its mean accuracy of 96.53% exceeded 60:20:20 by only 0.44 percentage points and 80:10:10 by 0.74 percentage points. Across three runs, approximate 95% t-based confidence intervals for mean accuracy were 94.95–97.23% for 60:20:20, 95.65–97.41% for 70:15:15, and 93.80–97.78% for 80:10:10. These intervals overlap extensively, and an exploratory one-way comparison across the nine runs yielded no statistically significant difference, $F(2,6) = 1.28$, $p = 0.345$. Thus, the ranking is compatible with sampling and optimization variability and could change under additional seeds or alternative partitions. The result is directionally consistent with evidence that training proportions can affect pretrained-model performance [14] and that allocating observations between model fitting and evaluation involves a measurable trade-off [15], but the present evidence supports a conditional, dataset-specific interpretation rather than a general recommendation.

Finite test-set uncertainty further qualifies the apparent advantage of 70:15:15. Based on the approximate test sizes generated by the three partitions, Wilson 95% confidence intervals for the best-run accuracy were 95.16–97.47% for 60:20:20, 95.27–97.83% for 70:15:15, and 94.61–97.86% for 80:10:10. Their substantial overlap shows that the best-run values cannot be treated as clearly distinct. The higher variation observed for 80:10:10 may partly reflect its smaller test subset, but this explanation is not empirically isolated. Each repetition simultaneously changed data allocation, sample composition, parameter initialization, mini-batch ordering, and stochastic augmentation, while the larger training subset also changed the information available for optimization. Consequently, the reported standard deviation represents combined run-to-run variability rather than a pure measure of split-ratio stability. A more conclusive design would hold the external test set fixed, repeat training across a larger number of controlled seeds, use bootstrap confidence intervals, and analyse partition and optimization effects separately.

The class-specific results are more consequential than the small aggregate differences. Perfect Late Blight precision indicates that predictions assigned to this class rarely produced false alarms, but recall of approximately 0.93–0.94 means that about 6–7% of actual Late Blight images were missed. For early disease screening, this is not a benign pattern because false negatives may delay intervention and allow a rapidly spreading infection to remain undetected. The model therefore should not be evaluated only through overall accuracy or weighted averages. Recall-oriented evaluation, class-specific confidence intervals, precision–recall curves, and cost-sensitive metrics are needed to represent the operational importance of missed Late Blight cases. Potential improvements include class-weighted loss functions, focal loss, threshold adjustment, targeted collection of difficult Late Blight stages, and augmentation designed around visually ambiguous lesions. Comparable compact-model research on rice and other crops likewise emphasizes the need to balance predictive strength, model simplicity, and deployment constraints [19], [20], [21], [22].

Using three repetitions is preferable to reporting a single favourable run because it exposes some degree of stochastic sensitivity. However, three observations per scenario provide a weak basis for estimating variance, checking distributional assumptions, or supporting inferential claims. The wide confidence intervals demonstrate that the apparent ordering remains uncertain. Moreover, random image-level stratification preserves class proportions but does not guarantee independence when near-duplicate images, repeated views of the same leaf, or shared backgrounds cross partitions. In repository-style datasets, such leakage can inflate performance by allowing the model to exploit source-specific visual regularities. Future experiments should incorporate duplicate screening and, where metadata permit, group-aware splitting by leaf, plant, acquisition session, field, or image source. Learning-curve analysis would also clarify whether additional training data genuinely improve generalization or whether performance has already plateaued.

Several limitations therefore constrain both scientific and practical interpretation. The study used one dataset, one architecture, three split ratios, and no independent external test set; it also lacked direct benchmarking against alternative lightweight networks and did not evaluate field robustness under variable illumination, cluttered backgrounds, partial occlusion, mixed disease stages, or device-specific imaging conditions. The principal implication is methodological rather than deployable: dataset allocation and repeated evaluation should be reported transparently because small changes in partitioning can alter both point estimates and uncertainty. Before the model can support farmer-facing early warning, future work should validate it on independently collected

multi-site images, prioritize Late Blight sensitivity, compare compact architectures under the same fixed-test protocol, and report complete efficiency measurements on actual target hardware.

VI. CONCLUSIONS

This study developed a MobileNetV3Small transfer-learning model for three-class potato leaf disease classification and examined the sensitivity of its performance to three training–validation–test allocations. Under the specific dataset, architecture, and training protocol evaluated, the 70:15:15 configuration produced the highest mean accuracy, the highest best-run performance, and the lowest observed accuracy variation across three repetitions. Its best run achieved 96.80% accuracy, while the mean accuracy reached 96.53% with a standard deviation of 0.35%. These results identify 70:15:15 as the most favourable option among the examined configurations, but they do not establish it as a universally optimal partition. The principal contribution is therefore methodological: dataset allocation should be treated as an empirical design decision and evaluated through repeated experiments, uncertainty reporting, and class-specific analysis rather than selected solely by convention or a single favourable run.

The practical significance of the model remains preliminary because the evaluation was conducted on a relatively uniform repository dataset rather than independently captured field images. Before deployment for farmer-facing early detection, future research should validate the experimental claim using a larger number of repeated stratified runs, a common untouched external test set, source-aware or group-aware splitting, and fixed-test-set learning curves to determine whether larger training allocations genuinely improve generalization. Further work should also compare additional lightweight architectures, report confidence intervals and statistical uncertainty, evaluate probability calibration and class-specific sensitivity—particularly Late Blight recall—and examine robustness under domain shift, background variation, illumination changes, partial occlusion, and device-specific imaging conditions. Source-aware external validation on independently collected multi-site field images should precede mobile or embedded implementation, while model size, computational cost, memory use, and inference latency should be measured on representative hardware before practical efficiency is claimed.

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